

UVSim Software Requirement Specifications

Functional Requirements:

1. The system shall load a user-selected file into memory, storing one instruction per location
2. The system shall execute instructions sequentially in ascending memory-location order, incrementing the program counter by one
3. The system shall terminate the program execution when it executes a HALT instruction
4. When the system executes a READ instruction, it shall prompt the user for a word and store the entered word at the operand memory location
5. When the system executes a WRITE instruction, it shall display the word stored at the operand memory location
6. When the system executes a LOAD instruction, it shall copy the word at the operand memory location into the accumulator
7. When the system executes a STORE instruction, it shall copy the accumulator value into the operand memory location
8. When the system executes a SUBTRACT instruction, it shall subtract the word at the operand memory location from the accumulator
9. When the system executes an ADD instruction, it shall add the word at the operand memory location to the accumulator
10. When the system executes a MULTIPLY instruction, it shall multiply the accumulator by the word at the operand memory location

11. When the system executes a DIVIDE instruction, it shall divide the accumulator by the word at the operand memory location
12. When the system executes a BRANCH instruction, it shall set the program counter to the operand memory location
13. The system shall display an error message and stop execution when a DIVIDE instruction attempts to divide by zero
14. The system shall display an error message when it encounters an instruction whose opcode is not defined in BasicML
15. The system shall display an error message and not run when the user selects a file that does not exist